

Md Nafiu Rahman

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About Me

I am a curious and driven computer science graduate and researcher with a keen interest in software engineering and LLMs. My long-term goal is to build intelligent tools that improve developer productivity and software reliability.

Education

Bangladesh University of Engineering & Technology (BUET), Dhaka *Feb 2020 – Mar 2025*
B.Sc. in Computer Science & Engineering CGPA: 3.81 / 4.00 (Final year: 3.94 / 4.00)
◦ Relevant coursework: Data Structures & Algorithms, Object Oriented Programming, Software Engineering, Database Systems, Numerical Methods, Machine Learning.

Notre Dame College, Dhaka *Jan 2017 – Jan 2019*
Higher Secondary Certificate GPA: 5.00 / 5.00

Bir Shrestha Noor Mohammad Public College, Dhaka *Jan 2009 – Dec 2016*
Secondary School Certificate GPA: 5.00 / 5.00

Work Experience

Summer Research Intern, University of Illinois Urbana-Champaign (UIUC), Urbana-Champaign, IL
May 2026 – Present
UIUC++ Summer Research in Software Engineering (SRSE) 2026
Supervisor: Dr. Shin Hwei Tan

Lecturer, Department of Computer Science and Engineering, BRAC University, Dhaka *July 2025 – Present*
◦ Teaching undergraduate courses including *Numerical Methods (CSE330)* and *Software Engineering (CSE470)*, delivering theory and laboratory sessions.
◦ Designing course materials, programming assignments, laboratory exercises, examinations, and coordinate lab activities while mentoring students in software engineering best practices.

Research Assistant, Department of Computer Science and Engineering, BUET, Dhaka *Mar 2025 – June 2025*
Supervisor: Dr. Anindya Iqbal
◦ Developed an LLM-based pipeline for automatically generating Playwright web testing scripts from high-level website descriptions and UI traces, evaluating the reliability of generated tests for practical software engineering workflows.

Publications

Conference Papers

[C1] **An Empirical Study on Calibration of LLMs in Code Generation.**
Authors: Mohammed Latif Siddiq, Md Nafiu Rahman, Joanna C. S. Santos.
(ICSME 2026 Research Track) [🔗](#)
Accepted at ICSME 2026 Research Track

[C2] **Secret Breach Detection in Source Code with Large Language Models.**
Authors: Md Nafiu Rahman, Sadif Ahmed, Zahin Wahab, Rifat Shahriyar, S. M. Sohan.
(ESEM 2025 Technical Track) [🔗](#)
Accepted at ESEM 2025 Technical Track

[C3] **Secret Leak Detection in Software Issue Reports using LLMs: A Comprehensive Evaluation.**
Authors: Sadif Ahmed, **Md Nafiu Rahman**, Zahin Wahab, Rifat Shahriyar, Gias Uddin.
([MSR 2026 Technical Track](#)) [🔗](#)
Accepted at MSR 2026 Technical Track

[C4] **IssueGuard: Real-Time Secret Leak Prevention Tool for GitHub Issue Reports.**
Authors: **Md Nafiu Rahman**, Sadif Ahmed, Zahin Wahab, Rifat Shahriyar, Gias Uddin.
([FSE 2026 Tool Demonstration Track](#)) [🔗](#)
Accepted at FSE 2026 Tool Demonstration Track

Workshop Papers

[W1] **BanglaForge: LLM Collaboration with Self-Refinement for Bangla Code Generation.**
Authors: Mahir Labib Dihan, Sadif Ahmed, **Md Nafiu Rahman**.
([BLP Workshop at AACL-IJCNLP 2025](#)) [🔗](#)
Accepted at BLP Workshop, AACL-IJCNLP 2025

Preprints

[P1] **A Survey on Agentic Security: Applications, Threats and Defenses.**
Authors: Asif Shahriar, **Md Nafiu Rahman**, Sadif Ahmed, Farig Sadeque, Md Rizwan Parvez.
([arXiv](#)) [🔗](#)
Submitted to TMLR

[P2] **Explainable Transformer-CNN Hybrid for Modeling Brain Aging from MRI Images.**
Authors: Wasif Jalal, **Md Nafiu Rahman**, Md Sohel Rahman.
([arXiv](#)) [🔗](#)
Submitted to Neural Networks

[P3] **EVCC: Enhanced Vision Transformer-ConvNeXt-CoAtNet Fusion with Adaptive Routing for Classification.**
Authors: Kazi Reyazul Hasan, **Md Nafiu Rahman**, Sadif Ahmed, Wasif Jalal, Shahriar Raj, Mubasshira Musarrat, Muhammad Abdullah Adnan.
([arXiv](#)) [🔗](#)

Technical Skills

Data Science & ML: Python, NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Torchvision.
Databases: PostgreSQL, PL/pgSQL, MongoDB, Firebase Firestore.
Full-Stack: Node.js (backend), HTML/CSS, React, Svelte, Flutter.
Languages: C/C++, Python, Java, JavaScript / TypeScript, PHP, Bash, Dart.

Language Proficiency

Bengali Native
English IELTS Overall 8.0 – Listening: 9, Reading: 9, Writing: 7, Speaking: 6.5

Achievements

- Top 20 finalists Robi Datathon 2024 (national deep learning competition).
- Deans list award and university merit scholarship at BUET.
- 5th at BLP 2025 Code Generation Challenge

Academic Projects

- **Machine Learning Algorithms and Neural Network from Scratch** – github.com/nafiurahman00/CSE-472 [🔗](#) – Implementations of core ML algorithms (logistic regression with ensembles, PCA/SVD for reconstruction, EM clustering) and a feed-forward neural network with Adam optimizer built from numpy.

- **Cryptography and Security Attacks** – github.com/nafiurahman00/CSE-406  – Implemented AES encryption and Diffie-Hellman key exchange, socket communication demos, and reviewed a mobile pentesting framework as part of course project.
- **Network Simulation** – github.com/nafiurahman00/CSE-322  – Implemented Congestion Control Algorithm, threaded server-client sockets, error correction algorithms and simulated wired/wireless mobility scenarios.
- **Operating System Internals with xv6** – github.com/nafiurahman00/CSE-314  – Implemented threading and synchronization primitives, system calls, and explored scheduler internals in xv6.
- **BusBuddy (Android)** – github.com/nafiurahman00/BusBuddy-Client-End  – Flutter app with Node.js backend, PostgreSQL and Firebase integration. Provided ticketing, schedules, tracking and real-time updates for university bus users.
- **Nishorgo (E-commerce)** – [Term-Project-2-2-Nishorgo](#)  – Full stack e-commerce site for plant sales with filtering, cart, admin analytics and order management.
- **Compiler (subset of C)** – github.com/nafiurahman00/Compiler  – Subset-of-C compiler using Lex/Yacc and 8086-style assembly generation: lexer, parser, and intermediate code generation.
- **Catch the Egg (Game)** – github.com/nafiurahman00/Catch-The-Egg  – OpenGL / Igraphics game for catching falling eggs; implemented game mechanics, scoring and difficulty scaling.

References

- **Dr. Rifat Shahriyar**, Professor, Department of CSE, Bangladesh University of Engineering and Technology (BUET) – rifat.shahriyar@gmail.com  – rifat@cse.buet.ac.bd 
- **S. M. Sohan**, Senior Engineer, Google – sohan39@gmail.com 
- **Dr. Md Rizwan Parvez**, Scientist, Qatar Computing Research Institute (QCRI) – rizwan@ucla.edu  – rizwan.incipient@gmail.com 
- **Dr. Farig Sadeque**, Associate Professor, Department of CSE, BRAC University – farig.sadeque@bracu.ac.bd 